PRACTICAL-6

Rendering External JSON Data using Fetch API, Search and Filter

Problem Definition : Display event lists, student profiles, notices, or FAQs from external JSON files using Fetch API. Implement dynamic rendering, search, filter, sorting, and pagination.

Code:

```
<!DOCTYPE html>
<html lang="en">

<head>

    <meta charset="UTF-8">
                <meta    name="viewport"    content="width=device-width,
initial-scale=1.0">

    <link rel="stylesheet" href="../css/style.css">
    <script src="../js/script.js"></script>

    <title>Student Profile</title>

</head>

<body>

<center>

    <br>

     <img src="../images/student hub.png" width="220" alt="Student Hub
Logo">

    <h1>STUDENT PROFILE</h1>

    <hr width="80%">

    <!-- Navigation -->

    <table border="1" cellpadding="10" cellspacing="0">

        <tr>
```

```
            <th><a href="../index.html">Home</a></th>

            <th><a href="dashboard.html">Dashboard</a></th>

            <th><a href="courses.html">Courses</a></th>

            <th><a href="attendance.html">Attendance</a></th>

            <th><a href="assignments.html">Assignments</a></th>

            <th><a href="results.html">Results</a></th>

            <th><a href="notice.html">Notice</a></th>

            <th><a href="contact.html">Contact</a></th>

            <th><a href="login.html">Logout</a></th>

        </tr>

    </table>

    <br>


    <fieldset style="width:900px;">

        <h2>Student Information</h2>

        <br>


        <!-- Search -->

        <input type="text"
               id="searchInput"
               placeholder="Search Student"
               onkeyup="searchStudent()">

        <br><br>
```

```
        <!-- Filter -->

        <select id="courseFilter" onchange="filterStudent()">

            <option value="all">All Courses</option>

            <option value="Computer Engineering">
                Computer Engineering
            </option>

        </select>

        <br><br>


        <!-- Sort -->

        <input type="button"
               value="Sort by Name"
               onclick="sortStudents()">

        <br><br>


        <!-- JSON Data will appear here -->

        <div id="studentData"></div>

        <br>

        <!-- Pagination -->

        <div id="pagination"></div>

        <br>


        <input type="button" value="Edit Profile">

        <input type="button" value="Download ID Card">
```

```
    </fieldset>


    <br>

    <hr width="80%">

    <footer>

        <p><b>© 2026 Student Hub Portal</b></p>

        <a href="contact.html">Contact</a> |

        <a href="#">Privacy Policy</a> |

        <a href="#">Help</a>

    </footer>

</center>

</body>

</html>
```

script.js:

```
// LOGIN
function login() {

    let username = document.getElementById("username").value;
    let password = document.getElementById("password").value;

    let usernamePattern = /^[0-9]{2}[A-Za-z]{3}[0-9]{3,4}$/;
                                    let         passwordPattern         =
/^(?=.*[A-Za-z])(?=.*[0-9])(?=.*@)[A-Za-z0-9@]{8,}$/;

    if (username == "" || password == "") {
        alert("Please enter username and password");
    }
    else if (!usernamePattern.test(username)) {
        alert("Invalid Username");
```

```
    }
    else if (!passwordPattern.test(password)) {
        alert("Invalid Password");
    }
    else {
        alert("Login Successful!");
        window.location.href = "dashboard.html";
    }
}


// PAGE LOAD
window.onload = function() {

// FAQ
let faq = document.createElement("button");
faq.innerHTML = "What is Student Hub?";
document.body.appendChild(faq);

let answer = document.createElement("p");
answer.innerHTML = "Student Hub helps students manage college
information.";
answer.style.display = "none";
document.body.appendChild(answer);

faq.onclick = function() {

    if (answer.style.display == "none") {
        answer.style.display = "block";
    }
    else {
        answer.style.display = "none";
    }
};

  // 3. NOTIFICATION BANNER
setTimeout(function() {

    let notification = document.createElement("div");

    notification.innerHTML = "🔔 New Notice: Check the Notice Board!";

    notification.style.backgroundColor = "#4a90e2";
```

```
    notification.style.color = "white";
    notification.style.padding = "10px";
    notification.style.margin = "15px";

    document.body.appendChild(notification);

}, 2000);

    // 4. HAMBURGER MENU
    let menuButton = document.createElement("button");
    menuButton.innerHTML = "☰ Menu";

    menuButton.style.position = "fixed";
    menuButton.style.top = "10px";
    menuButton.style.right = "190px";
    menuButton.style.backgroundColor = "#4a90e2";
    menuButton.style.color = "white";

    document.body.appendChild(menuButton);

    let navigation = document.querySelector("table");

    navigation.style.display = "none";

    menuButton.onclick = function() {

        if (navigation.style.display == "none") {
            navigation.style.display = "table";
        }
        else {
            navigation.style.display = "none";
        }
    };


    // 5. DARK MODE
    let darkButton = document.createElement("button");
    darkButton.innerHTML = "Dark Mode";

    darkButton.style.backgroundColor = "#4a90e2";
    darkButton.style.color = "white";
    darkButton.style.position = "fixed";
    darkButton.style.top = "10px";
```

```
    darkButton.style.right = "100px";

    document.body.appendChild(darkButton);

    darkButton.onclick = function() {

        document.body.style.backgroundColor = "black";
        document.body.style.color = "white";
    };


    // 6. LIGHT MODE
    let lightButton = document.createElement("button");
    lightButton.innerHTML = "Light Mode";

    lightButton.style.backgroundColor = "#4a90e2";
    lightButton.style.color = "white";
    lightButton.style.position = "fixed";
    lightButton.style.top = "10px";
    lightButton.style.right = "10px";
    document.body.appendChild(lightButton);
    lightButton.onclick = function() {
        document.body.style.backgroundColor = "#dff3ff";
        document.body.style.color = "black";
    };
};
// PRACTICAL 6

let students = [];

fetch("../data/student.json")
.then(response => response.json())
.then(data => {

    students = data;
    display(students);

});


function display(data) {

    let table = "<table border='1' cellpadding='10'>";
```

```
    table += "<tr>";
    table += "<th>Name</th>";
    table += "<th>Enrollment</th>";
    table += "<th>Course</th>";
    table += "<th>Semester</th>";
    table += "</tr>";

    for(let i = 0; i < data.length; i++) {

        table += "<tr>";
        table += "<td>" + data[i].name + "</td>";
        table += "<td>" + data[i].enrollment + "</td>";
        table += "<td>" + data[i].course + "</td>";
        table += "<td>" + data[i].semester + "</td>";
        table += "</tr>";

    }

    table += "</table>";

    document.getElementById("studentData").innerHTML = table;
}


function searchStudent() {

                                        let          text          =
document.getElementById("searchInput").value.toLowerCase();

    let result = students.filter(function(student) {

        return student.name.toLowerCase().includes(text);

    });

    display(result);
}


function sortStudents() {

    students.sort(function(a,b) {
```

```
        return a.name.localeCompare(b.name);

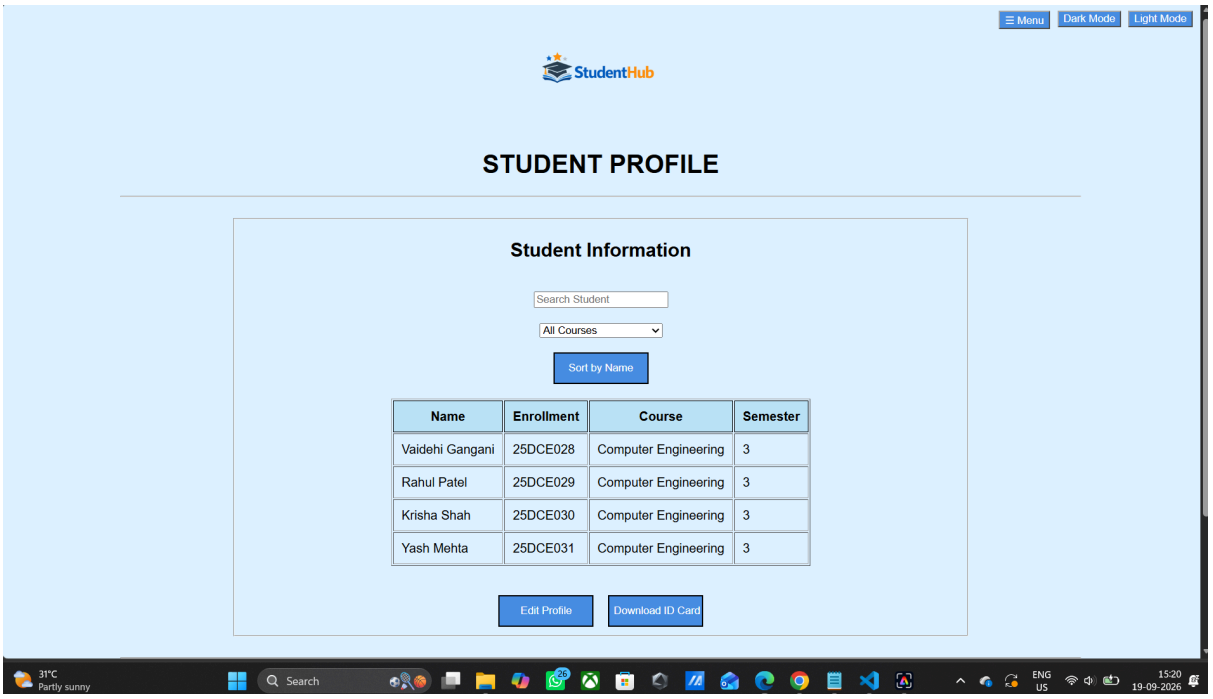
    });

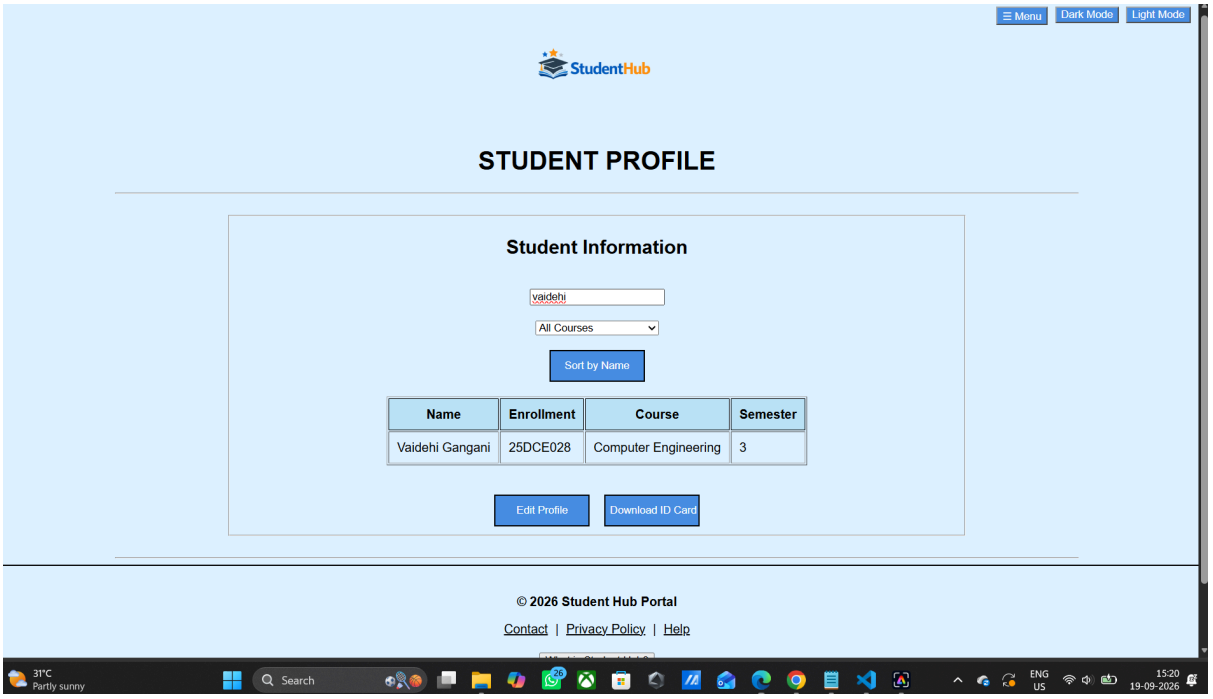
    display(students);
}
```

Student.json:

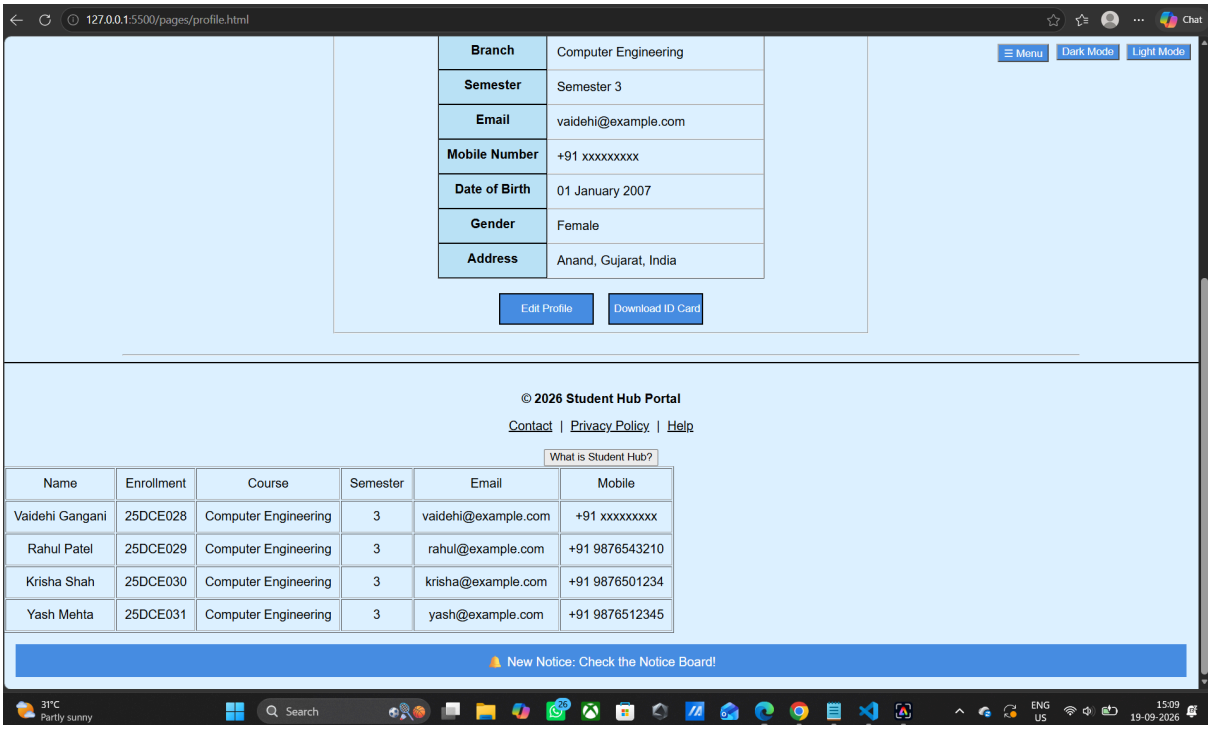
```
[
    {
        "name": "Vaidehi Gangani",
        "enrollment": "25DCE028",
        "course": "Computer Engineering",
        "semester": 3,
        "email": "vaidehi@example.com",
        "mobile": "+91 xxxxxxxxx"
    },
    {
        "name": "Rahul Patel",
        "enrollment": "25DCE029",
        "course": "Computer Engineering",
        "semester": 3,
        "email": "rahul@example.com",
        "mobile": "+91 9876543210"
    },
    {
        "name": "Krisha Shah",
        "enrollment": "25DCE030",
        "course": "Computer Engineering",
        "semester": 3,
        "email": "krisha@example.com",
        "mobile": "+91 9876501234"
    },
    {
        "name": "Yash Mehta",
        "enrollment": "25DCE031",
        "course": "Computer Engineering",
        "semester": 3,
        "email": "yash@example.com",
        "mobile": "+91 9876512345"
    }
```

```
]
```


☰ Menu
Dark Mode
Light Mode
StudentHub
STUDENT PROFILE
Student Information
Search Student
All Courses
Sort by Name
Name
Enrollment
Course
Semester
Vaidehi Gangani
25DCE028
Computer Engineering
3
Rahul Patel
25DCE029
Computer Engineering
3
Krisha Shah
25DCE030
Computer Engineering
3
Yash Mehta
25DCE031
Computer Engineering
3
Edit Profile
Download ID Card
31°C
Partly sunny
Search
ENG
US
15:20
19-09-2026


☰ Menu
Dark Mode
Light Mode
StudentHub
STUDENT PROFILE
Student Information
vaidehi
All Courses
Sort by Name
Name
Enrollment
Course
Semester
Vaidehi Gangani
25DCE028
Computer Engineering
3
Edit Profile
Download ID Card
© 2026 Student Hub Portal
Contact | Privacy Policy | Help
31°C
Partly sunny
Search
ENG
US
15:20
19-09-2026

Challenges



Problem Faced

Initially, the output was not displayed properly. The student table was appearing at the bottom of the page instead of inside the Student Information section

Reason

This happened because the JavaScript was displaying the JSON data in the studentData div, but the existing page structure and CSS were not properly arranged for the dynamically generated table. Also, the JavaScript file was running on other pages where the studentData element was not present.

Mistake in Code

I had added the JSON rendering code in the common script.js file, which is loaded on multiple pages. Because of this, the student table was generated separately at the bottom of the profile page.

Correction

I checked the HTML structure and JavaScript execution and corrected the placement of the JSON data section so that the student data is displayed in the required section.